



**Andukuri Ravindra**

S/o A. Sri Hari Rao (Late),  
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Visakhapatnam – 09

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randukur@gitam.edu

## ABOUT ME

Research-oriented mechanical engineering professional with extensive experience in CAD/CAM, simulation, intelligent manufacturing, and decision-making systems. Actively engaged in research, teaching support, industrial consultancy, and scholarly publishing in the areas of smart manufacturing, AI-based decision modeling, and sustainable engineering.

## SKILLS

### WORK / SOFTWARE

|             |             |
|-------------|-------------|
| SOLIDWORKS  | <div></div> |
| CATIA       | <div></div> |
| FUSION 360  | <div></div> |
| AUTOCAD     | <div></div> |
| ANSYS       | <div></div> |
| ADAMS       | <div></div> |
| MATLAB      | <div></div> |
| PYTHON      | <div></div> |
| 3D PRINTING | <div></div> |
| MS OFFICE   | <div></div> |
| PAGEMAKER   | <div></div> |
| PHOTOSHOP   | <div></div> |

# Andukuri Ravindra

## ACADEMIC EXPERIENCE

(2009 – present)

### Senior Technician

[Dept. of Mechanical Engineering, GIT, GITAM (Deemed to be University), Visakhapatnam]

Designed and guided the students' projects, instructed CAD/CAM software to the students, prepared all engineering drawings 3D modeling using SolidWorks, CATIA, Fusion 360, AutoCAD, etc, and prepared 3D models for 3D printing. Also designed banners, brochures, invitations, newsletters, and other media according to theme and topic for each brand. Assistance with administrative-related work.

## INDUSTRY EXPERIENCE

(1995 – 2009)

- Worked as a **Design Engineer** in the Naval Science & Technological Laboratory (NSTL), Visakhapatnam, under the project of AUCE (12th Nov '04 to 17th July '09).
- Worked as a **Draughtsman** in M/s. NAVARC Consultants and Engineers, Visakhapatnam. (From 04.03.02 to 27.08.04)
- Worked as a **Customer Support Engineer** in Octal Computer Services, Visakhapatnam (April 2001 to July 2001)
- Worked as a **Quality Control Supervisor** in M/s. Allen Reinforced Plastics Pvt., Ltd., Hyderabad, during the years 1996 to 1997.
- Worked as a **Production Supervisor** in M/s. VISWAKIRTI Engg. , Pvt. Ltd. Hyderabad, during the years 1995 to 1996.

## EDUCATION

**Master of Business Administration (MBA – HRM)**, pursuing (4<sup>th</sup> Semester) in Andhra University Online and Distance Education.

**M.Tech. (Industrial Engineering)**, First Class, 2024, Andhra University, Visakhapatnam

**B.E (Mechanical)**, First Class, 2014, Andhra University, Visakhapatnam

- Diploma in Mechanical Engineering**, 1995, Govt. Polytechnic, Visakhapatnam.
- Intermediate (MPC)**, 1991, Board of Intermediate Education, Hyderabad, A.P
- S.S.C**, 1989, Board of Secondary Education, Hyderabad, A.P

## TECHNICAL SKILLS

- MSC Multibody Dynamics Certification Course**, DIYguru (NEAT–AICTE), Successfully Completed, October 2025.
- Completed the NPTEL Online certificate course entitled “**Research Methodology**” organized by Indian Institute of Technology, Madras. (Jul-Sept 2025).

## PERSONAL

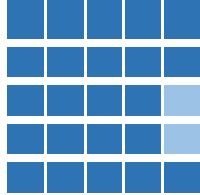
COMMUNICATION

ORGANIZATION

TEAM PLAYER

CREATIVITY

SOCIAL



- Completed the NPTEL Online certificate course entitled **"Product Design and Development"** organized by Indian Institute of Technology, Roorkee. (Jul-Aug 2023).
- Internship on **"Electrical Vehicle Design Master Class"** from Pantech Solution Ltd, February 2022.
- Training in **"AutoDesk Fusion 360"** from ISCT, AutoDesk authorized academic partner, January 2022.
- Training in **"ANSYS"** from Central Institute of Tool Design (CITD), Sub-Center, Visakhapatnam. (07.08.2013 to 06.09.2013)
- Training in **"CATIA V5"** from Indo-German Institute of Advanced Technology (IGIAT), Visakhapatnam. (28.07.2008 to 19.09.2008)
- Training in **"Pro-Engineer (Pro-E 2000i & Wildfire)"** from Millenium Software Solutions, Visakhapatnam. (02.03.2005 to 10.06.2005)
- **"Diploma in computer Maintenance"** from Octal computer Services, Visakhapatnam - 15th March 2001
- Training in **"CAD/CAM with CNC programming"** from Central Institute of Tool Design (CITD), Hyderabad. (01.03.99 to 30.06.1999)
- Job Training in **"Auto CAD R-14"** at Naval Science and Technological Laboratory, Visakhapatnam. (01.02.98 to 30.06.98)
- Training in **"AutoCAD"** from Institute of Modern Technology, Visakhapatnam - May 1998

## PROGRAMS ATTENDED

Year 2025

- Presented a technical paper titled **"Design and Optimization of Bio-Inspired Green Materials using AHP-GRA and Finite Element Simulation"** at AICTE-ATAL-VAANI National Conference on Digital Makeover in Engineering Applications: Innovation, Technology and Sustainability, held during 16-17 December 2025, at Government Polytechnic, Visakhapatnam.
- Presented / Co-authored a technical paper titled **"Simulation-Driven Digital Twin for Condition Monitoring of Rotary Machinery under Industry 4.0"** at the National Conference on Condition Monitoring (NCCM-2025), jointly organized by Gas Turbine Research Establishment (GTRE), Bengaluru and Condition Monitoring Society of India (CMSI), held during 27-28 November 2025, at HAL Management Academy, Bengaluru.
- Participated and successfully completed the **One-Week Faculty Development Program on "Advanced Design and Manufacturing using Autodesk Fusion"** organized by the Department of Mechanical Engineering, Chaitanya Bharathi Institute of Technology (A), Hyderabad, held during 04-09 August 2025.
- Participated in the **Workshop on Autodesk Fusion Fundamentals**, organized by Autodesk Authorized Academic Partner (Design Labs), held on 17 June 2025, at GITAM College, Visakhapatnam.

Year 2023

- Participated in International Conference on **"Emerging Aspects of Manufacturing, Thermal and Design Engineering (MATHED-22)"** organised by the Department of Mechanical Engineering, National Institute of Technology Hamirpur during February 15-17, 2023.

"I am constantly developing my technical skills, keeping in view the rapidly evolving technical aspects. As part of this, I have the technical skills to demonstrate my theoretical and practical knowledge. I have the ability to design and develop mechanical components."

Year 2022

- Participated and completed the AICTE Training And Learning (ATAL) Academy Online Elementary FDP on "**Machinery Condition Monitoring**" from 10/01/2022 to 14/01/2022 at Zakir Husain College of Engineering & Technology. (Score: 60%)

Year 2021

- Participated and completed the AICTE Training And Learning (ATAL) Academy Online Elementary FDP on "**3D Printing & Design**" from 06/12/2021 to 10/12/2021 at NIT Warangal. (Score: 86%)
- Participated and completed the AICTE Training And Learning (ATAL) Academy Online Elementary FDP on "**3D Printing & Design**" from 08/11/2021 to 12/11/2021 at University college of engineering, Osmania university. (Score: 68%)
- Participating in "**3-Day Online Workshop On Computational Fluid Dynamics (CFD) Using Ansys Fluent**", organized by the School of Mechanical Engineering (SMEC), VIT-AP University, Amaravati (AP), India during 18th to 20th November, 2021.
- Participating in "**2-day International Conference on Condition Monitoring**" jointly organized by CMSI and TATA Steel, Jamshedpur during 21-22, January 2021

Year 2020

- 2nd One Week International Faculty Development Program – ONLINE on "**Role of Materials and Processing in Additive Manufacturing: 3D Printing to Industry-2020**", from 19th - 24th July 2020 organized by Department of Mechanical Engineering, GLA University, Mathura in association with Indian Institute of Metals (IIM), Mathura Chapter.
- Completed eLearning program on "**Bearings**" organized by TATA Steel on 22nd May 2020
- Completed eLearning program on "**Machine Learning**" organized by TATA Steel on 2nd May 2020
- Completed eLearning program on "**Industry 4.0**" organized by TATA Steel on 24th April 2020

Before 2020

- Five day Short term course on "**Advances in CAD/CAM**" organized by Department of Mechanical Engineering, NIT Rourkela on 21-25 May 2018.
- Five day training programme on "**Effective Office Management and Advanced Computer Skills**" conducted by GIT, GITAM University on 20-24 May 2013.
- One day training programme on "**Web-based e-Financial Monitoring Report (e-FMR) and Financial Management Training**" under TEQIP II, conducted by NPIU-New Delhi and SPFU-AP at

"In this rapidly evolving technological era, teaching and learning are considered as vital role. I learn by attending the programs to improve my technical knowledge. Attending faculty development programs, conferences, workshops, and seminars have helped me to expand my technical skills"

University College of Engineering, Osmania University, Hyderabad on 15th May 2013.

- Two day training programme on “**Academic Activities & Management Information Systems**” under TEQIP II, conducted by NPIU-New Delhi and SPFU-AP at University College of Engineering, Osmania University, Hyderabad on 13-14 May 2013.

**Till this date, successfully completed the 14 Technical courses offered by COURSERA.**

*“ I am assisting and organizing the various department's activities in order to study and enhance program management for the benefit of both the department and the Institute”*

## ORGANISED PROGRAMS

- Program Coordinator in Guest lecture on “**Applications of Artificial Intelligence for Mechanical Engineers**” conducted by Dept. of Mechanical Engineering GST, GITAM University 20<sup>th</sup> July 2022.
- Team Member in “**Design Competition**” conducted by Dept. of Mechanical Engineering GST, GITAM University with AutoDesk, USA during 25<sup>th</sup> to 27<sup>th</sup> May 2022.
- Co-convenor in 2-day workshop on “**3D Printing And Additive Manufacturing**” conducted by Dept. of Mechanical Engineering GIT, GITAM University during 11-12 June 2018
- Organizing member in “**National Conference on Materials & Manufacturing -2017**” conducted by Dept. of Mechanical Engineering GIT, GITAM University during 22-23 Dec 2017
- Organizing member in “**International Conference on Condition Monitoring-2016**” conducted by Dept. of Mechanical Engineering GIT, GITAM University
- Organizing member in “**International Conference on Condition Monitoring-2011**” conducted by Dept. of Mechanical Engineering GIT, GITAM University.

## ROLE IN ADMINISTRATION

*“Not only am I working with technical work, but I am also assisting with administrative work in order to give constant assistance to the department and the Institute”*

- Administration and presentation work for the NAAC Peer Committee Visit in 2023 as an **Effective member, Mechanical Engineering of NAAC Institutional Committee** and was Accredited with “A++” Grade.
- Worked as Technical assistant, “**Technical Education Quality Improvement Programme Phase –II (Worth of Rs. 4 Crores), Subcomponent 1.2**”, Principal Office, GIT, GITAM University from Jan 2011 to Aug 2017.
  - Supported to the Procurement Management service and Office Administrative Work for completion of TEQIP – II, Subcomponent 1.2 Project (SPFU/NPIU).
- Administration and presentation work for the NAAC Peer Committee Visit during 19- 21 February on 2017 as an **Effective member, Mechanical Engineering of NAAC Institutional Committee** and was Accredited with “A+” Grade. (Received the appreciation from Institution for my contribution)

## CONSULTANCY PROJECTS

- Completed consultancy project, Supporting staff, “ **(i) Modelling, analysis and design for transfer trolley system (ii) Capacity calculation of winch and rope**” as offered by M/s. Geo Marine Dynamics (I) Pvt. Ltd, Visakhapatnam worth Rs. 1.8 lakhs+GST as per Institute norms (2020)

"As a part of my involvement in research and consulting initiatives, I have aided and conducted consultation work. The primary tasks in consultant work are design, modelling, and report" generation"

- Preparation of Modelling and design for transfer trolley system by using CATIA, Capacity calculation of winch and rope
- Completed Consultancy project, Technical Support, **"FEM Analysis of Machinery Foundation Structures"** as offered by Reveille Allied Engineers Pvt. Ltd.(Indian Navy), Ship building Company, Visakhapatnam worth of Rs. 3.00 lakhs (2015-16)
  - Prepared 3D models of FIFTEEN onsite existing Machinery test beds at Shipbuilding Centre, Visakhapatnam in AutoCAD and CATIA.

## PATENT

- **Indian Patent Published (2025)**  
System and Method for Adaptive Neuro-Fuzzy Decision-Making for Intelligent Process Planning and Predictive Maintenance  
**Application No.:** 202541084081 | **Published:** 24-10-2025 | **RQ Filed:** 08-01-2026  
**Status:** Application Awaiting Examination (IPO, India)
- **Indian Patent Published (2024)**  
*An Integrated Thermal Energy Storage System for a Helmet*  
**Application No.:** 202441049738 | **Published:** 12-07-2024 | **RQ Filed:** 17-06-2025  
**Status:** Application Awaiting Examination (IPO, India)

## PUBLICATIONS

### JOURNALS / ARTICLES

"As a Author / co-author, I've contributed to the publication of research publications and Books/Monographs. These are the outcomes of my technical abilities and expertise. I hope that many more publications/books will be produced to help my career and the careers of my colleagues, faculty members, and others."

- Andukuri, Ravindra, and V. Naga Malleswari. **"CNN-GNN-based feature recognition and fuzzy MCDM strategy ranking from STEP files: a case study on intelligent machining decisions."** Expert Systems with Applications 303 (March 2026): 130618. (Q1-Scopus). <https://doi.org/10.1016/j.eswa.2025.130618>
- Andukuri, R., Sankaranarayana, K. **Enhanced tribological performance of PTFE-reinforced acetal polymer composites: an experimental and statistical study.** emergent mater. (February 2025). (Q2-Scopus). <https://doi.org/10.1007/s42247-025-01014-x>
- Andukuri, R., & Rao, C. M. Operations Research Forum **Application of Fuzzy CODAS for the Optimal Selection of Condition Monitoring Equipment in Industrial Rotating Machinery.** (2024, September). (Q2 - Scopus). <https://doi.org/10.1007/s43069-024-00376-y>
- Ravindra et.al. International Journal of Multidisciplinary Research (IJFMR) – Research Article entitled **"Enhanced Condition Monitoring of Gearboxes using Fuzzy comprehensive evaluation and group decision making: A case Study"** July-August 2024 (Google Scholar).
- Ravindra et.al. International Journal of Engineering Research & Technology (IJERT) – Research Article entitled **"Enhancing reliability of cryogenic globe valves using Fuzzy AHP – TOPSIS integrated FMEA methodology"** July 2024 (Google Scholar).
- Ravindra et.al. International Journal of Engineering Research & Technology (IJERT) – Research Article entitled **"Evaluation of Automotive Materials for Electric Vehicles using CODAS and ELECTRE methods with sensitivity analysis"** July 2024 (Google Scholar).
- Ravindra et. al. Journal of the Institution of Engineers (India): Series D. – Research article entitled **"Analysis of corrosion Polarization behavior in GNP / TiB2 reinforced hybrid Al-Mg-Si-MMNCs for marine environment"** May 2024 (Q3 – Scopus).



- ✦ Ravindra et al. Progress in Additive Manufacturing, Springer Link - Research Article entitled "**Automatic feature recognition from STEP file for smart manufacturing**" Volume 1, Issue 21, March 2024 9 <https://doi.org/10.1007/s40964-024-00583-3>. (Q1 – Scopus).
- ✦ Ravindra et al, International Journal of Engineering Research & Technology - Research paper entitled "**An Experimental Study of the TiB2-Reinforced AA7075 Manufactured by the Stir Casting Technique**" Volume 11, Issue 1, January – 2022 (Google Scholar)
- ✦ Ravindra et al, Springer Journal paper entitled "**Experimental Investigation on Mechanical Characterization of Nanoclay-Reinforced Banana Fiber/E-Glass/Epoxy Resin Hybrid Nanocomposite**", Recent Advances in Material Sciences, Volume/Issue 1 (2019), pp 609-625, Lecture Notes on Multidisciplinary Industrial Engineering, DOI:10.1007/978-981-13-7643-6\_50 (Web of Science)
- ✦ Ravindra et al, Elsevier Journal Paper entitled "**Water absorption kinetics and mechanical properties of ultrasonic treated banana fiber reinforced-vinyl ester composites**", Procedia Materials Science, Volume/Issue 5C, (2014) pp. 311-315, <https://doi.org/10.1016/j.mspro.2014.07.272> (Web of Science)

## BOOKS / CHAPTERS/MONOGRAPHS

### Books Published

- ✦ **Sustainable Material Selection for Instrument Panel Using the Combinative Distance-Based Assessment Method in Multi-Criteria Decision-Making Techniques**

Ravindra Andukuri, Dr. Kesava Rao V.V.S., Dr. Prasad V.V.S.  
Walnut Publication | ISBN: 978-9359113777 | Published: Jan 2025

Link: <https://walnutpublication.com/books/>

- ✦ **20 High-Scoring Mechanical Engineering Project Ideas with Industry-Ready Insights: Projects Focused on Mechanical Design, Simulation, AI/ML, MCDM, and Sustainability**

Ravindra Andukuri

Amazon Kindle Direct Publishing | ASIN: B0FBKK5RTP | Published: May 2025

Link: <https://www.amazon.in/dp/B0FBKK5RTP>

"Several academic books published through international publishers with ISBN registration."

- ✦ **Optimization Of Additive Manufactured Lattice Structure** published by Lambert Academic Publications, Germany, ISBN No. 978-620-4-98305-9, 2022
- ✦ **Rotor Dynamic Performance In Turbo Machinery Refurbishment** published by Lambert Academic Publications, Germany, ISBN No. 978-620-4-97901-4, 2022  
Translated and published in **7 international languages** [German, ISBN: 978-620-4-92981-1; Spanish, ISBN: 978-620-4-92992-7; French, 978-620-4-92993-4; Italian, ISBN: 978-620-4-92997-2; Portuguese, ISBN: 978-620-4-93001-5; Russian, ISBN: 978-620-4-93005-3]
- ✦ **CI Engines - Performance and Emission** published by Lambert Academic Publications, Germany, ISBN No. 978-620-4-95301-4, 2022  
Translated and published in **7 international languages** [German, ISBN: 978-620-4-81137-6; Spanish, ISBN: 978-620-4-81138-3; French, 978-620-4-81139-0; Italian, ISBN: 978-620-4-81140-6; Portuguese, ISBN: 978-620-4-81141-3; Russian, ISBN: 978-620-4-81142-0]
- ✦ **Nano Lubricant Performance in Automotive Engines** published by Lambert Academic Publications, Germany, ISBN No. 978-620-4-97810-9, 2022

**LinkedIn :**

<https://www.linkedin.com/in/andukuri-ravindra-58434aa>

**ORCID :**

<https://orcid.org/0000-0001-5619-9942>

**Scopus Author ID:**

59294090800, [58307516600](#)

**Research Gate :**

<https://www.researchgate.>

"Recognized reviewer for top journals, member of ASME & IAENG, life member of CMSI, committed to professional development and excellence "

- ✦ **Mechanical Characterization of Hybrid laminated Composites** published by Lambert Academic Publications, Germany, ISBN No. 978-613-9-95339-4, 2018
- ✦ **Marine Propeller Noise Control (Modelling and Analysis)** published by Lambert Academic Publications, Germany, ISBN No. 978-613-9-86317-4, 2018
- ✦ **Shell-and Tube Heat exchanger: Design and Maintenance** published by Lambert Publications with ISBN No. 978-3-8465-0588-5, 2013

## AWARDS / RESEARCH RECOGNITION

- ✦ **Best Paper Award**  
Awarded for the paper titled "Design and Optimization of Bio-Inspired Green Materials Using AHP-GTMA and Finite Element Simulation" at the **AICTE-ATAL-VAANI National Conference on Digital Makeover in Engineering Applications: Innovation, Technology and Sustainability (DMEATS-2K25)**, held on **16-17 December 2025** at Government Polytechnic, Visakhapatnam.
- ✦ ORCID ID: 0000-0001-5619-9942 | Scopus Author IDs: 59294090800, 58307516600.  
Actively serving as a **peer reviewer** for international journals, including:
  - *Expert Systems with Applications*
  - *Journal of Intelligent Manufacturing*
  - *Journal of the Brazilian Society of Mechanical Sciences and Engineering*
  - *Journal of the Institution of Engineers (India), Series D*
  - *National Academy Science Letters*

## PROFESSIONAL INSTITUTIONS

- ✦ **Member**, American Society of Mechanical Engineers (ASME), elected by the Board of Governors in 2025, contributing to professional activities related to mechanical engineering research and practice.
- ✦ **Member**, International Association of Engineers (IAENG), since March 2021.
- ✦ **Life Member**, Condition Monitoring Society of India (CMSI), since 2009, with professional engagement in condition monitoring, vibration analysis, and predictive maintenance.

**Declaration:**

All the information/details mentioned above are true to the best of my knowledge and beliefs.



**(A. RAVINDRA)**